MUHAMMAD **FAREED**

Lahore, Pakistan | mfareedkhan012@gmail.com | +92 303 4756731

Website | Google Scholar | GitHub | LinkedIn | **ORCID:** 0009-0005-9509-2108

Research Interests

Medical vision-language models and multimodal clinical AI; efficient foundation-model adaptation (PEFT/QLoRA); medical image–text generation and evaluation; trustworthy AI; low-resource and multilingual NLP.

Education

- Master of Science (MS) in Data Science

2024 - 2026

Riphah School of Computing & Innovation (RSCI), Riphah International University, Islamabad — **Lahore Campus**

CGPA: 4.0/4.0 | **First position** in the graduating cohort | **Merit Scholarship** 2024-2026

- **Thesis:** Retrieval-Augmented, Parameter-Efficient Fine-Tuning of Vision–Language Model for Chest X-ray Report Generation.
- **Supervisor:** Dr. Jamal Uddin | **Published in Scientific Reports** | Paper

- Bachelor of Education (B.Ed) Secondary

2021 - 2024

Virtual University of Pakistan — **Lahore, Pakistan**

- Bachelor of Science (Honors) in Mathematics

2012 - 2016

Division of Science & Technology, University of Education, Township Campus — **Lahore, Pakistan**

Publications & Manuscripts | Google Scholar

Published – First Author

[1] **Fareed, M.**, Uddin, J., Sayyid, A. H., Ahmed, A., & Sattar, M. A. (2026). "CXRG-SVLM: A Parameter-Efficient Small Vision-Language Model for Multi-View Chest X-ray Report Generation in Resource-Constrained Environments". *Scientific Reports (Nature), SCIE, JCR Q1, JIF: 4.9* | **Paper**

[2] **Fareed, M.**, Fatima, M., Uddin, J., Ahmed, A., & Sattar, M. A. (2025). "A Systematic Review of Ethical Considerations of Large Language Models in Healthcare and Medicine". *Frontiers in Digital Health, JCR Q1, JIF: 5.2, Web of Science (ESCI), CiteScore: 6.8* | **Paper**

Under Review

[3] **Fareed, M.**, Uddin, J., Nauman, M. A., Khan, N., Billah, M., & Sattar, M. A. (2026). LSTM-Based Deep Learning Models in Healthcare and Medicine: A Systematic Review. *PLOS ONE.*

Submitted & In Preparation

[4] Ahmed, A., Riaz, S., & **Fareed, M.** (Nov 2026). "PEFT-mBERT: Benchmarking a Transformer-Based Encoder via Low-Rank and Activation-Based Adaptation for Resource-Efficient Punjabi Sentiment Analysis". *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP). (submitted)*

[5] Mir, S. S., Uddin, J., & **Fareed, M.** (2026). "Hybrid CNN–Transformer Based Diabetic Retinopathy Detection with Explainable AI Using Multi-Dataset Fundus Images". (in preparation)

Research Experience

Graduate Researcher – MS Thesis

2024 - 2026

Riphah International University, Lahore | **Advisor:** Dr. Jamal Uddin

CXRG-SVLM: Parameter-Efficient Multi-View Chest X-ray Report Generation

- Developed and evaluated a **medical vision-language model for multi-view chest X-ray report generation**, combining a frozen RAD-DINO medical image encoder with a 4-bit Qwen2.5-3B language model and rank-16 **QLoRA adaptation**.
- Implemented **attention-based multi-view fusion**, a **32-token visual prefix**, clinical-indication conditioning, and contrastive visual–text alignment with **38.1M trainable parameters (1.91%)**.
- Led experiments, manuscript preparation, and report evaluation using lexical, semantic, radiology-specific, and entity-level measures, analyzed omissions, unsupported findings, and limitations requiring external and clinical validation.

First-Author Healthcare AI Research

2025 - 2026

Trustworthy and Ethical Large Language Models in Healthcare | **Advisor:** Dr. Madeeha Fatima

- Systematically reviewed studies on **ethical & trustworthy AI issues in healthcare LLMs**; examined **privacy, bias, transparency, and accountability**; and developed a four-layer framework covering **regulatory & technical safeguards, human oversight, and accountability**.

Collaborative Research *2026*

Medical image analysis and parameter-efficient NLP | **Advisor:** Dr. Jamal Uddin & Dr. Saman Riaz

- Contributed to research design, experiments, and **explainable multi-dataset diabetic-retinopathy analysis** using a hybrid CNN–Transformer, and contributed to research on **parameter-efficient multilingual BERT adaptation for low-resource languages**, including research design, literature analysis, and manuscript development.

Professional & Teaching Experience

Nov 2026 - Present Working as **Visiting Lecturer – Computer Vision** at Riphah International University, Lahore, AI Campus.
- Teaching Computer Vision to BS Artificial Intelligence (BSAI) undergraduate students.

Mar 2018 - Present Working as a **Science & Mathematics Teacher (PST, BPS-14)** at GES Attari Saroba, Lahore | School Education Department, Government of Punjab.

Sep2026 - Nov2026 Worked as a **Management Trainee Officer — AI/ML** at Excellence Delivered (ExD), Lahore.

2012 - 2023 Earlier roles: **Mathematics/Physics Instructor and Tutor**; digital marketing & web development trainer (Lahore)

Selected Technical Projects | GitHub

Co-team lead, Institute of Advanced Engineering (IAE) technical projects associated with Stanford Engineering CGOE's 'Innovation-Driven Product Development Using AI' program. | January 2026

- Roman Urdu Conversational Language Model: Developed an instruction fine-tuning pipeline using *Llama-3.1-8B, Unsloth, and 4-bit QLoRA*, with qualitative evaluation of generated responses under limited-compute conditions.

- Length-Controlled News Summarization: Adapted *4-bit Mistral-7B-Instruct using LoRA* for length-controlled summarization; evaluated outputs using ROUGE, BERTScore, length adherence, and qualitative assessment of unsupported content.

- Multilingual Sentiment Detection: Adapted *Gemma-3-270M* for sentiment classification, exploring lightweight language-model fine-tuning.

Technical Skills

Programming & Frameworks	Python, LaTeX, PyTorch; TensorFlow; Keras; Hugging Face Transformers; Unsloth.
Methods	LoRA/QLoRA; PEFT; 4-bit quantization; transfer learning; CNNs; Transformers
Cloud, Data & Deployment	AWS; Microsoft Azure; Google Cloud Platform (GCP); Docker; Kubernetes; SQL and NoSQL databases; Apache Hadoop; Apache Spark; Tableau; Power BI; Microsoft Excel.

Selected Certifications & AI Training

Innovation-Driven Product Development Using AI	Stanford Engineering CGOE x PMYP	Feb 2026	Certificate
Deep Learning Specialization (completed all five courses)	DeepLearning.AI	Dec 2025	Certificate
Data Science Ethics	University of Michigan	Sep 2025	Certificate
Generative AI Engineering & Fine-tuning Transformers	IBM	Jul 2025	Certificate

Honors, Awards & Scholarships

- **Top 1% Merit-Based Scholarship** (50% throughout MS program)**,** Riphah International University, 2024-2026.
- **Certificate of Merit Achievement, Dean's Honor List** — MS Data Science, Riphah School of Computing & Innovation.
- Awarded the **Innovation-Driven Product Development Using AI** certificate by Stanford Engineering CGOE, Feb 2026.
- **Best Teaching Performance Award**, Progressive Education Network, Lahore, December 2024.

Research Presentations

9 Dec 2025 **Three Minute Thesis Competition (3MT)** — International Conference on Computing & Innovation (ICCI 2025); presenter, certificate of participation.

Sep 2025 **Research Week** — RSCI, Poster presentation of MS thesis research.

Leadership & Community Service

2018 - 2020 Worked as **Vice President** at Tameer Kadah Foundation (NGO), Lahore.
- Led operations, fundraising, mobility-aid drives, and free medical camps to empower children with disabilities.

References

References available upon request.